

- 9 (a) Electrons in a vacuum are accelerated through a potential difference of 84 kV. The electrons then strike a metal target and X-rays are produced.
- (i) Calculate the minimum wavelength of the X-rays that are produced.

wavelength = m [2]

- (ii) The melting points of two metals are given in Table 9.1.

Table 9.1

metal	melting point/°C
copper	1090
tungsten	3420

Suggest why the metal target is made from tungsten rather than copper.

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 [2]

- (b) An X-ray beam is incident normally on a sample of soft tissue and bone as shown in Fig. 9.1.

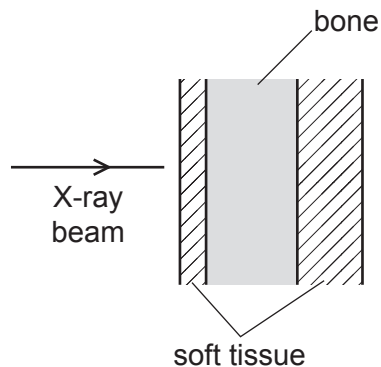


Fig. 9.1

Data for the two materials are given in Table 9.2.

Table 9.2

medium	linear attenuation coefficient μ/cm^{-1}	specific acoustic impedance $Z/10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
soft tissue	0.22	1.7
bone	3.0	7.8

The total thickness of soft tissue is x . The total thickness of bone is also x .

The incident intensity of the X-ray beam is I_0 . The transmitted intensity of the X-ray beam is 13% of the incident intensity.

Determine x , in cm.

$x = \dots\dots\dots \text{ cm [3]}$

- (c) (i) Define the specific acoustic impedance of a medium.

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 [2]

- (ii) Use data from Table 9.2 to calculate the percentage of the intensity of ultrasound that is transmitted at a boundary between soft tissue and bone.

percentage transmitted =% [2]

- (iii) The ultrasound is now incident on the sample of soft tissue and bone shown in Fig. 9.1.

Suggest **two** reasons why the transmitted intensity through the sample is less than the answer in (c)(ii).

1

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2

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[2]

[Total: 13]